



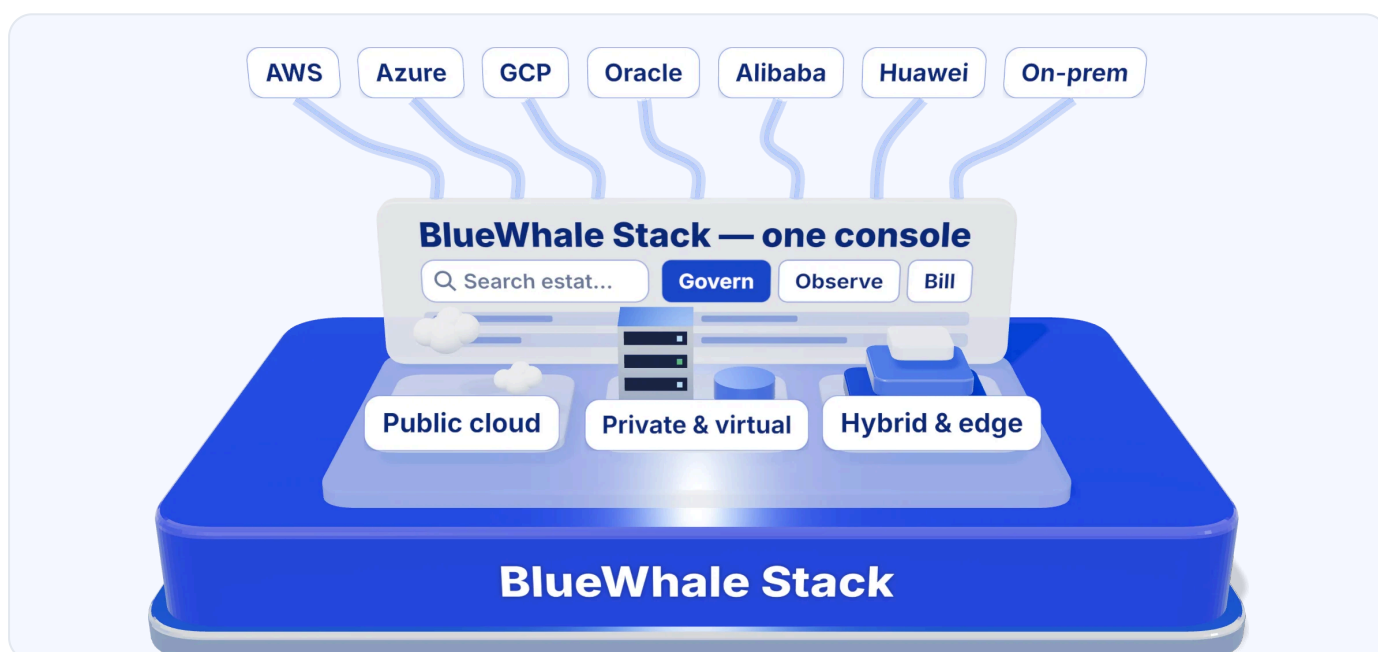
BLUEWHALE STACK

WHITEPAPER

Platform capability guide

Fourteen modules, nine families — what each does, how it works, and what is live today

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Every module of the platform in one document: summary, proof points, capabilities with their mechanisms, how it works, and availability by edition.

Fourteen modules ship across the nine capability families today. This guide states, for each module, what it does, how it works, the proof points and its current maturity — GA, beta, preview or in progress — in the same words the product pages use.

The families

Management & Delivery

Provisioning, landing zones, blueprints and day-2 operations — delivery with governance built in.

Whalenomics · FinOps

Budgets, forecasts, chargeback and continuous optimization — spend decomposed to workload, department or tenant.

Security & Identity One identity fabric (SAML/OIDC), least-privilege access, continuous scanning — uniform posture everywhere.	Governance & Audit Policy as configuration; controls monitored continuously; the auditor's report generated on demand.
Whale AI — incl. offline AI for operations, documentation and compliance — your choice of model, able to run fully inside the perimeter.	Migration & Discovery Inventory, dependency mapping and rehearsed waves with rollback — for entry, movement between estates, and exit.
Observability & ITSM Health, events and service operations — integrated with ServiceNow, Jira and existing tooling.	Tenancy & Monetization Native multi-tenancy, catalogs, marketplace, per-tenant metering and billing feeds.
Sovereign Operations In-country deployment, segregation, air-gapped classes and offline-tolerant edge.	

Cloud Connectors — Generally available

Cloud Connectors are how an estate joins the platform: read-only, least-privilege credentials for each public cloud, and an Edge Agent for private and virtualised estates that connects outbound over HTTPS so credentials never leave your site. Six public clouds are live today; hybrid and sovereign stacks connect through the same agent.

6 Public clouds live — AWS, Azure, GCP, Oracle, Alibaba, Huawei	Read-only IAM role, app registration or service account — never write access by default	Outbound only Edge Agent uses HTTPS out; no inbound firewall rules
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Capabilities

Least-privilege cloud access AWS connects with an IAM role and external ID, Azure with an Entra app registration holding the Reader role, GCP with a Viewer service account, Oracle with an API signing key, Alibaba and Huawei with read-only access keys. Write actions are granted separately, per policy.	Edge Agent for private estates A small agent per site inventories VMware, Hyper-V, Nutanix and OpenShift through their own APIs and reports over outbound-only HTTPS. Hybrid and sovereign stacks — Azure Stack, Huawei Cloud Stack, Alibaba Apsara Stack — attach the same way.
Connector health Every connector shows last sync, scope and errors; a failing credential raises an ITSM incident rather than silently going stale.	Business and ITSM connectors ServiceNow and Jira integrations for tickets, plus CRM/ERP connectors for ownership and cost context; some business connectors are still in beta.

How it works

- 1

Create credentials in the cloud
Follow the guided steps for each provider — the console shows the exact role, policy or key to create, scoped to read.
- 2

Register the connector
Paste the role ARN, app credentials or key into the connector form; the platform validates access and runs a first discovery.
- 3

Watch it stay healthy
Sync status, scope changes and expiring secrets are surfaced on the health page and as incidents, so nothing quietly drops out of the inventory.

Service Catalog — Generally available

The Service Catalog lets teams request approved cloud resources without touching cloud consoles. AWS provisioning is live for EC2, S3, RDS, VPC and EFS, with Azure and GCP catalog items, approval workflows, governance guardrails and Whale AI sizing advice built into the request — so self-service and control stop being a trade-off.

AWS live EC2 · S3 · RDS · VPC · EFS provisioned from the catalog	Guardrails Approved sizes, regions and tags enforced before submit	Whale AI Sizing recommendations inside the request form
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Capabilities

Governed self-service Teams pick from catalog items you approve — with sizes, regions, tags and naming already constrained — instead of being handed console access.	Approval workflows Route requests by cost, environment or resource type to the right approver; low-risk items can auto-approve, everything is recorded.
Whale AI sizing The request form asks Whale AI for a right-sized recommendation grounded in your inventory and cost history, before a single instance is oversized.	Audit trail Who requested, who approved, what was provisioned and where — every catalog action lands in the platform audit log and the ITSM change record.

How it works

- 1

Publish catalog items
Define the resources teams may request and the guardrails that apply to each.
- 2

Request and approve
A developer selects an item, gets a sizing suggestion, submits; the approval policy decides who signs off.

3 Provision and track

The platform provisions with its own credentials, tags the resource, and adds it to inventory and cost tracking automatically.

Landing Zone Builder — Beta

Landing Zone Builder is a visual designer for compliant cloud foundations — AWS Control Tower, Azure Cloud Landing Zone and GCP foundations — that generates the multi-account baseline as Terraform HCL with guardrails, tags and account structure built in. Design the foundation as a diagram; get the code that stands it up.

AWS · Azure · GCP Control Tower, CLZ and GCP foundation patterns	Baseline HCL Multi-account, versioned, reviewable	Guardrails Policy, tagging and network controls from day one
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Capabilities

Visual foundation design Lay out organisations, accounts, networks and shared services on a canvas that mirrors each provider's reference architecture.	Guardrails as configuration Choose the controls — SCPs and policies, tagging standards, network baselines — and they are generated into the code, not documented beside it.
Generated multi-account baseline The output is Terraform for the whole foundation, ready for review and your pipeline; WhaleForge can extend it.	Industry patterns Banking, telco, healthcare and government starting points that reflect segregation and residency requirements.

How it works

1 Design

Pick the provider pattern and draw the account and network structure.

2 Select guardrails

Turn on the controls the estate must have; the designer validates the combination.

3 Generate

Export the baseline HCL and apply it through your pipeline.

WhaleForge IaC — Beta · deployment runner coming

WhaleForge lets platform teams write infrastructure in a plain YAML DSL and compiles it to real Terraform HCL for AWS, Azure and GCP — with live HLD, LLD and TOGAF diagrams that stay in sync with the code, PDF export for architecture reviews and Git import for existing repositories. The deployment runner is the next milestone.

YAML → HCL Real Terraform output, not a proprietary runtime	HLD · LLD · TOGAF Diagrams generated from the same source	Beta Available to Enterprise & up on request
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Capabilities

A readable DSL Describe networks, compute, data and policies in YAML that architects and reviewers can read; WhaleForge expands it into provider-specific Terraform.	Diagrams that cannot drift HLD, LLD and TOGAF views are rendered from the source on every change and exported to PDF for design authorities and audits.
Git import Bring existing Terraform repositories in, see them as diagrams, and evolve them in the DSL where it helps.	Industry landing-zone packs Starting points for Banking, Telco, Healthcare and Fintech that encode guardrails, tags and account structure from day one.

How it works

- 1

Author
Write or import the definition; validation runs as you type.
- 2

Compile and review
Generate Terraform and the diagrams; review both in the same change.
- 3

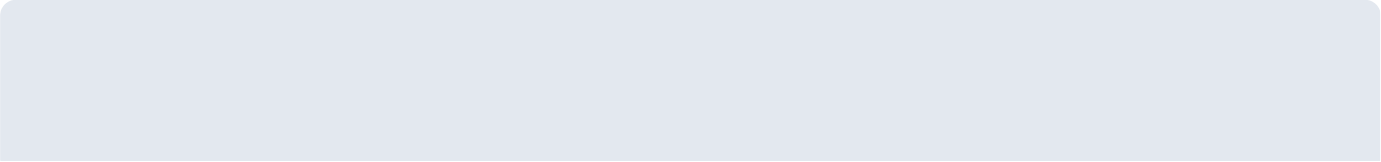
Apply
Run through your existing pipeline today; the WhaleForge deployment runner is coming next.

Whalenomics — Whale AI use cases live · full backend in progress

Whalenomics is the FinOps family: budgets, forecasts, chargeback and continuous optimisation, with spend decomposed to workload, department or tenant. Today the Whale AI FinOps use cases — cost narratives, anomaly explanations, rightsizing and commitment advice — run on live billing data from AWS, Azure and GCP; the full Whalenomics backend for budgets and chargeback is in progress.

12 Whale AI FinOps use cases live today	AWS · Azure · GCP Pricing maps and billing ingestion	Per tenant Chargeback and metering model (in progress)
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Capabilities



Cost visibility with ownership Billing records are joined to the inventory, so every dollar resolves to a workload, an owner and a tag — the basis for showback and chargeback.	Anomalies and explanations Whale AI flags unusual spend and explains it in plain language against the inventory: what changed, where, and who owns it.
Rightsizing and commitments Idle and oversized resources, storage tiers and commitment coverage are surfaced with the saving and the action — from the inventory, not a separate tool.	Budgets, forecasts, chargeback Budgets per unit with forecasts and alerts, and chargeback into your ERP or an operator's BSS — the Whalenomics backend currently in progress.

How it works

- 1

Ingest and join
Billing exports from each cloud are normalised and joined to inventory and tags.
- 2

Explain
Whale AI answers 'why did this rise' with the resources behind the number.
- 3

Decide and allocate
Rightsize, commit, and allocate cost to the unit that owns it; budgets and forecasts follow.

Identity & Access — Generally available

Identity & Access is the platform's identity fabric: federate the identity providers you already run — Entra ID, Okta, Auth0, AWS Identity Center, Google IAM, OneLogin, on-prem Active Directory and the cloud IAMs — provision users automatically over SCIM, and apply one role model across every connected cloud. Enterprise adds Whale IAM-PAM for privileged sessions.

9+ Identity providers over SAML 2.0 and OIDC	SCIM 2.0 Automatic user and group provisioning	One RBAC Roles mapped from IdP groups, enforced on every cloud
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Capabilities

Single sign-on SAML 2.0 and OIDC federation with Microsoft Entra ID, Okta, Auth0, AWS Identity Center, Google IAM, OneLogin and on-premises AD, with step-by-step setup guides for each.	Auto-provisioning SCIM 2.0 creates, updates and deactivates users as your directory changes, and maps IdP groups to platform roles — joiners and leavers are handled where HR already handles them.
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Role-based access, cross-cloud Roles such as viewer, operator and approver apply across AWS, Azure, GCP and on-prem estates at once, with scope by account, workload or tenant.	Privileged access (Enterprise & up) Whale IAM-PAM brokers privileged sessions with approval gates and recording; the Government Edition makes PAM and MFA mandatory on every role.
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How it works

- 1

Federate
Register BlueWhale Stack as an application in your IdP and exchange SAML or OIDC metadata — a few minutes per provider.
- 2

Map groups to roles
Choose which directory groups become which platform roles, and the scopes those roles cover.
- 3

Provision automatically
Turn on SCIM so accounts and roles follow the directory; access reviews and the audit trail come for free.

Cloud Audit & Evidence — Included in Enterprise & up

Cloud Audit & Evidence treats policy as configuration and monitors the controls behind it continuously across every estate, so evidence is generated all the time instead of assembled before an inspection. Reports are produced on demand for the board, the auditor and the regulator, with obligations mapped per regime — DPDP, GDPR, RBI, CERT-In, SEBI and more.

Continuous Controls monitored, not sampled quarterly	Per regime Obligations mapped to platform controls	On demand Examiner-grade reports as parameters, not projects
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Capabilities

Policy as configuration Residency, encryption, access, logging and tagging policies are defined once and applied across clouds; drift is a finding, not a surprise.	Continuous control monitoring Each control is evaluated against the live inventory and telemetry; failures open findings with the resources and owners attached.
Regime mapping Frameworks such as India DPDP, GDPR, RBI outsourcing, CERT-In and SEBI CSCRf are mapped to controls, so one evidence set answers several regulators.	Examiner-grade reporting Generate the report an inspection asks for — scope, controls, evidence, exceptions — from a system of record with a tamper-evident trail; the Government Edition adds WORM-backed audit logs.

How it works

- 1

Declare
Choose the regimes that apply and the policies the estate must meet.
- 2

Monitor
Controls are checked continuously; findings route to owners through ITSM.
- 3

Report
Produce the evidence pack for a period, a regime or an estate in minutes.

Whale AI — Generally available

Whale AI is an intelligence layer woven into every module rather than a chatbot beside them: 50+ production use cases across cost, security, ITSM, operations, migration and governance, each grounded in your live inventory, billing, tickets and findings. It runs in three tiers — Spark, Tide and Abyss — with your choice of model, including fully offline inside the perimeter for sovereign estates.

50+ Use cases, each with declared data dependencies	Spark · Tide · Abyss Three tiers routed by task complexity	Offline In-region and air-gapped models for sovereign deployments
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Capabilities

Grounded, not generative guesswork Every use case declares the providers it reads — inventory, cost, tickets, findings — and cites them in the answer; numbers come from the platform, not the model.	Tiered intelligence Spark answers interactive questions in under three seconds; Tide reasons over migration plans and IaC; Abyss runs multi-step agents for the hardest cross-module work.
In the flow of work An assistant drawer in every module, streamed answers, exports to PDF and Excel, and prompt caching so repeat questions are instant.	Inside the perimeter Bring your model or run ours in-region; sovereign deployments run Whale AI fully offline with zero external data exposure.

How it works

- 1

Ask, in context
From any screen, ask about what you are looking at — a workload, a bill, an incident.
- 2

Ground
The use case gathers the live facts it declares, scoped to your permissions.
- 3

Answer with citations
The right tier reasons over the facts and streams an answer with the evidence behind it.

Inventory & Discovery — Generally available

Inventory & Discovery builds one live map of every resource you run — across AWS, Azure, Google Cloud, Oracle, Alibaba and Huawei, and on-premises VMware, Hyper-V, Nutanix and OpenShift estates connected through the Edge Agent. Resources are grouped by workload with their dependencies, so platform, security and finance teams work from the same picture instead of six consoles and a spreadsheet.

6 + on-prem Public clouds plus private estates in one inventory	Workload Grouping with dependency mapping, not raw resource lists	PDF · XLSX · CSV Exports for audits, CMDB and finance
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Capabilities

Continuous discovery Connectors sync each account on a schedule and on change events; the Edge Agent inventories hypervisors over outbound-only HTTPS. New resources appear without anyone registering them.	Workload grouping and dependencies Resources are normalised into one model and grouped into workloads (VPC → subnet → compute → database), so an application is a unit you can search, cost and migrate — not a list of instance IDs.
Live lifecycle actions Start, stop, tag, resize or retire from the inventory itself, with role-based approval where policy requires it and every action written to the audit log.	Reports and feeds Export the estate as PDF, Excel or CSV, and feed the same inventory to Whalenomics (cost), Cloud Audit & Evidence (controls) and the Migration Engine (waves).

How it works

- 1 Connect**
 Add cloud accounts with read-only credentials and register Edge Agents for on-prem sites. The first discovery typically completes within minutes of a connector going green.
- 2 Normalise**
 Every provider's resource types are mapped to one schema with tags, owners, regions and relationships — the shared model every other family reads.
- 3 Work from one map**
 Search, filter by provider or workload, act on resources, and export — the inventory is the system of record for the estate.

Migration Engine — Assessment live · execution hooks in progress

The Migration Engine industrialises entry, movement and exit: on-premises workloads discovered by the Edge Agent are auto-classified, scored with a 6R assessment for cost, effort and blockers, and planned into rehearsable waves with rollback and Whale AI guidance. Assessment and wave planning are live; execution hooks are in progress.

6R Rehost · replatform · refactor · repurchase · retire · retain, scored	Waves Dependency-aware planning with rollback	Exit-ready The same engine rehearses regulator-mandated exit plans
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Capabilities

Auto-classification Discovered workloads are classified by type, dependencies and data gravity, so the assessment starts from facts rather than interviews.	6R assessment Each workload gets a recommended strategy with estimated target cost, effort and blockers — and the evidence behind the score.
Wave planning with rollback Waves are cut along dependency boundaries, rehearsed, and carry a rollback plan; Whale AI drafts the sequence and flags risks.	Execution hooks (in progress) Handoff to tooling for the move itself, with status flowing back into the wave board and ITSM changes.

How it works

- 1

Discover
Edge Agents inventory the source estate and map dependencies.
- 2

Assess
6R scoring with cost, effort and blockers per workload.
- 3

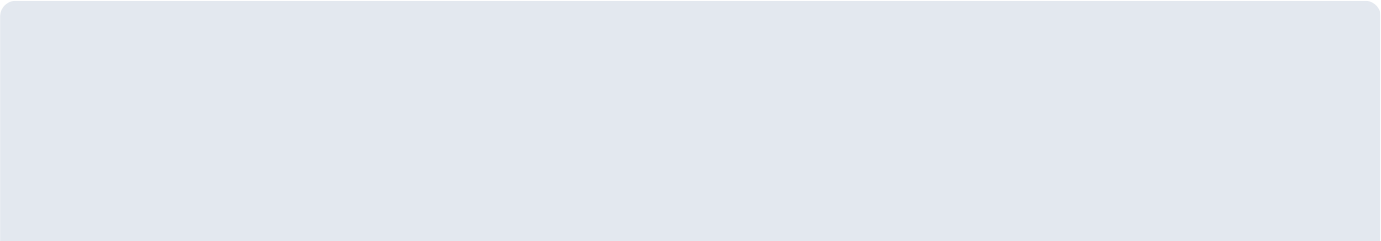
Plan waves
Sequence, rehearse, approve; execute with rollback when the hooks land.

Observe — Generally available

Observe bundles production-grade observability into the platform — logs, metrics and traces in one place, SLOs with burn-rate alerting, synthetics, usage metering and notification channels — so teams get full observability of the estate without procuring and stitching a separate monitoring contract.

Logs · metrics · traces One data model, one query surface	SLOs Error budgets with burn-rate alerts	Bundled Included in Standard & up — no separate contract
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Capabilities



Unified telemetry Ingest logs, metrics and traces from cloud services and agents into one store, correlated with the inventory so a signal always resolves to a workload and an owner.	SLOs and burn-rate alerts Define availability and latency objectives per service; alerts fire on error-budget burn rate, not raw thresholds, which cuts noise and pages the right team.
Synthetics and metering Probe endpoints from your regions, and meter usage per tenant or business unit — the same numbers Whalenomics and Tenancy use for billing.	Notification channels Route alerts to email, chat and on-call tools, and open ITSM incidents automatically for SLO breaches.

How it works

- 1

Ingest
Cloud-native telemetry via provider APIs; agents and OpenTelemetry for applications; the Edge Agent for on-prem.
- 2

Define objectives
Set SLOs per service with the platform suggesting baselines from observed behaviour.
- 3

Alert and act
Burn-rate alerts route to channels and ITSM; dashboards and traces are one click from the incident.

ITSM — Generally available

ITSM is service management built for cloud operations: incidents with P0–P4 SLAs and breach detection, change and problem management, a knowledge base, a CMDB fed by the live inventory and kanban boards — with tickets raised directly from cloud events and two-way integration with ServiceNow and Jira where those remain the system of record.

P0–P4 SLA classes with breach detection and escalation	CMDB Configuration items kept in sync with the inventory	ServiceNow · Jira Two-way ticket integration
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Capabilities

Incidents from the estate A failed connector, an SLO burn, a policy violation or a cost anomaly opens an incident with the affected workload attached — no copying from one tool to another.	Change and problem management Changes carry the approval trail from the Service Catalog; problems link recurring incidents to a root cause and a fix.
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CMDB without manual upkeep Configuration items are the inventory — when a resource changes, its CI changes; relationships come from dependency mapping.	Integrations Push and sync tickets with ServiceNow and Jira, keep your existing queues and on-call, and use the platform as the source of cloud context.
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How it works

- 1

Classify
Set SLA classes and escalation paths per service; the estate's events map to them automatically.
- 2

Work the queue
Incidents, changes and problems in kanban or list views, each linked to the live resources involved.
- 3

Close with evidence
Every state change is recorded — the audit trail doubles as the evidence the regulator and the post-incident review both need.

Tenancy & Monetization — Preview · GA Q4 2026 (Telco & Datacenter Edition)

Tenancy & Monetization is the Digital Experience Layer for operators: native multi-tenancy with isolation and quotas, white-label portals, service catalogs and a marketplace, and per-tenant metering that feeds your own OSS/BSS and invoicing — so a telco or datacenter operator publishes governed cloud services under its own brand and bills for them. In preview now, general availability targeted for Q4 2026.

Native Multi-tenancy with isolation and quotas per tenant	White-label Portals, catalog and marketplace under your brand	OSS/BSS Metering and billing feeds into your existing stack
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Capabilities

Tenant isolation and quotas Each tenant gets its own identity scope, resources, quotas and policies on shared operator capacity, enforced by the platform rather than by convention.	White-label experience Portals with your logo, colours and domain; a catalog of your services and a marketplace for partners and ISVs.
Metering and billing feeds Usage is metered per tenant and pushed to your BSS or invoicing — the operator keeps its commercial machinery and adds cloud products to it.	SLA and tenant health Per-tenant SLA reporting and health visibility drive renewals and support, with ITSM tickets scoped to the tenant.

How it works

- 1

Model tenants
Define tenants, quotas and policies on the operator's estate.
- 2

Publish services
Build the catalog and brand the portal; open the marketplace to partners.
- 3

Meter and bill
Usage flows to OSS/BSS; SLAs and health are visible per tenant.

Sovereign Operations — Generally available — Government Edition

Sovereign Operations makes sovereignty a property of the architecture: in-country deployment with directorate-level segregation under central policy, air-gapped deployment classes with an offline update channel, FIPS-validated cryptography, always-on privileged access management, a WORM-backed audit log and offline-tolerant edge sites — with Whale AI operating entirely inside the perimeter.

Air-gapped Deployment classes with offline updates and no call-home	FIPS · PAM · WORM Validated crypto, always-on PAM, immutable audit	In-region AI Whale AI Abyss runs on local models only
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Capabilities

Segregation under central policy Ministries, directorates or agencies operate their own estates with autonomy inside national guardrails — one policy plane, many isolated domains.	Air-gapped classes The most sensitive estates run fully disconnected with an offline update channel; hybrid classes connect to sovereign clouds such as AWS GovCloud, Azure Government and Google Distributed Cloud.
Hardened by default FIPS-validated cryptography, MFA mandatory on every role, PAM with session recording always on, residency enforced and evidenced.	Offline-tolerant edge Remote sites keep operating under headquarters-grade policy when disconnected and reconcile when they reconnect.

How it works

- 1

Classify estates
Assign each estate a deployment class — in-country connected, sovereign cloud, or air-gapped.
- 2

Segregate and govern
Directorates get isolated domains; policy, identity and audit remain central.
- 3

Evidence continuously
State-audit evidence and compliance packs (FedRAMP / IRAP / StateRAMP-style) are generated from the WORM-backed trail.

Module availability by edition

Module	Standard	Enterprise	Telco & Datacenter	Government
Cloud Connectors	Included	Included	Included	Included
Service Catalog	Included	Included	Included	Included
Landing Zone Builder	—	Included	Included	Included
WhaleForge IaC	—	Included	Included	Included
Whalenomics	Included	Included	Included	Included
Identity & Access	Included	Included	Included	Included
Cloud Audit & Evidence	—	Included	Included	Included
Whale AI	Included	Included	Included	Included
Inventory & Discovery	Included	Included	Included	Included
Migration Engine	—	Included	Included	Included
Observe	Included	Included	Included	Included
ITSM	Included	Included	Included	Included
Tenancy & Monetization	—	—	Included	—
Sovereign Operations	—	—	—	Included

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